How the code is organized

Open Factory Studio is a Qt 6 Widgets program made of a few libraries:

Target	Kind	What it does
ofs	program	Starts the app, loads the plugins and shows the main window
ofsshell	static library	The main window, menus, project tabs and the program's dialogs
ofscore	shared library	The base of everything: settings, actions and shortcuts, themes, Lua presets, projects, the shared editors (code, sprites, animations, sounds, palettes), drawing and audio playback
ofsformats	shared library	File formats and codecs (SFF, SND, PCX, ACT, FNT, GIF and so on). Plain C++20 without Qt
ofslua	shared library	Lua 5.4, shared by the core and all plugins
ofs_mugen	plugin	Everything for M.U.G.E.N: project types, the syntax database, its editors and tools, Run in engine and the emulator

ofs uses ofsshell, which uses ofscore. ofscore uses ofsformats and ofslua (through sol2). The M.U.G.E.N plugin only talks to ofscore.

Folders

Folder	What's in it
src/formats	File formats without Qt: util, image, codecs, palette, sff, snd, wav and fnt
src/core	The core library
src/shell	The main window and dialogs
src/app	The program's entry point and resources
src/engines/<name>	Engine plugins
data	Shared data: presets, icon sets, styles, color sets, docs and the engines' own data
tests	Unit, GUI, corpus and oracle tests. tests/fixtures makes the test files in code
tools	Developer tools (ofs-sffdump and some Python scripts)
third_party	Small libraries kept in the repository (LodePNG)
docs	These documents

At runtime the program finds its data folder through OFS::Paths (src/core/app/Paths.h). User settings are kept in %LOCALAPPDATA%\Open Factory Studio on Windows, or the usual place on other systems.

Main ideas

- **Engines** (core/engine): an engine plugin says which project types, versions and file types it handles, and creates the editors for them. See the Engine plugins document.
- **Actions** have fixed names like file_open, spr_add or anim_play. Menus, toolbars, panels, shortcuts and presets all refer to actions by name.
- **Presets** are Lua scripts that build the toolbars and panels from named controls.
- **Projects** own their files, and every file has its own undo history. Text files keep their exact text, so the parts you didn't touch are saved byte for byte.
- **File formats** (OFS::Formats) read files with bounds checks and return errors in Result<T> values instead of throwing. The documents (SffDocument, SndDocument, FntDocument) keep the original bytes of everything you didn't edit, so saving an unchanged file gives the same file back. The byte layouts are in the File formats document.